

OpenCPU SDK

LBS API Notes

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Notice

This document is specifically for **N706B** and **N520** modules.

This document is intended for system engineers (SEs), development engineers, and test engineers.

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About This Document

Scope

This document is applicable to N706B and N520 modules.

Audience

This document is intended for [system engineers \(SEs\)](#), [development engineers](#), and [test engineers](#).

Change History

Issue	Date	Change	Changed By
1.0	2023-05	Initial release	Niu Sulong

Conventions

Symbol	Indication
	This warning symbol means danger. You are in a situation that could cause fatal device damage or even bodily damage.
	Means reader be careful. In this situation, you might perform an action that could result in module or product damages.
	Means note or tips for readers to use the module

1 Overview

Location-Based Service (LBS) is a technology that determines the location of a mobile device using wireless communication base stations.

The LBS API enables access to base station location data and Wi-Fi location information. This document explains the related LBS APIs to help developers quickly implement relevant applications.

2 Data Structure

(1) Callback Function

```
typedef void (*nwy_loc_cipgsmloc_callback) (nwy_lbs_result_t *text);
```

(2) Callback parameters nwy_lbs_result_t

```
typedef struct{
    NWY_LBS_STATUS_E result;
    union
    {
        nwy_cipgsmloc_info_t data;//
        char errmsg[255];//
    }info;
}nwy_lbs_result_t;
```

Parameter type	Note
NWY_LBS_STATUS_E result;	(1) NWY_LBS_STATUS_SUCCESS: Read response nwy_cipgsmloc_info_t. (2) NWY_LBS_STATUS_RESP_ERROR: Read errmsg for details of the error.
nwy_cipgsmloc_info_t data	Response data
char errmsg[255]	Error message, containing an HTTP response message.

```
typedef struct
{
    double lat;//Latitude, in degrees
    double lng;//Longitude, in degrees
    double alt; //Estimated accuracy of the location, in meters
}nwy_cipgsmloc_info_t;
```

Parameter type	Note
double lat	Latitude, in degrees.
double lng	Longitude, in degrees.
double alt	Estimated accuracy of the location, in meters.

(3) NWY_LBS_STATUS_E: LBS Status

```
typedef enum
{
    NWY_LBS_STATUS_SUCCESS,// Request successful, read response nwy_cipgsmloc_info_t
    NWY_LBS_STATUS_PKT_ERROR,
    NWY_LBS_STATUS_BUSY,
    NWY_LBS_STATUS_NET_ERROR,
```

```
NWY_LBS_STATUS_SIM_STATUS_ERROR,  
NWY_LBS_STATUS_TIMEOUT,  
NWY_LBS_STATUS_SEND_FAIL,  
NWY_LBS_STATUS_RECV_FAIL,  
NWY_LBS_STATUS_CONNECT_FAIL,  
NWY_LBS_STATUS_DNS_FAIL,  
NWY_LBS_STATUS_RESP_ERROR, // Response error, read `errmsg` for details  
NWY_LBS_STATUS_UNKNOWN_ERROR  
}NWY_LBS_STATUS_E;
```

(4) Data testing:

The API uses the Amap database. Location accuracy can be tested using the Amap location picker:

3 APIs

This interface function definition can be found in `nwy_lbs_api.h`.

3.1.1 nwy_lbs_cip

Format	<code>nwy_error_e nwy_lbs_cip(int cid ,nwy_loc_cipgsmloc_callback cb);</code>
Description	Base station location request.
Parameter	Input parameter: cid: Dialing channel information (1-7). cb: Callback status check interface. See section 3.1 for details.
Return Value	Success: <code>NWY_SUCCESS</code> Failure: Other enumeration values
Remarks	• Dialing must be initiated. • IMEI must be valid. • The following website can be used to locate the current position based on the obtained latitude and longitude and to check the deviation (Gaode Map): https://lbs.amap.com/tools/picker

3.1.1 nwy_lbs_wifi

Format	<code>nwy_error_e nwy_lbs_wifi(int cid ,nwy_wifi_scan_list_t *scan_list,nwy_loc_cipgsmloc_callback cb);</code>
Description	WiFi location request.
Parameter	Input parameter: cid: Dialing channel information (1-7). scan_list: The obtained Wi-Fi scan data, acquired using the interface <code>nwy_wifi_scan</code> from the header file <code>nwy_wifi_api.h</code> . cb: Callback status check interface. See section 3.1 for details.
Return Value	Success: <code>NWY_SUCCESS</code> Failure: Other enumeration values
Remarks	• Dialing must be initiated. • IMEI must be valid. • The following website can be used to locate the current position based on the obtained latitude and longitude and to check the deviation (Gaode Map): https://lbs.amap.com/tools/picker